MULTIMEDIA SYSTEMS

Assignment 3 – Answers (Simple Notes)

Q1. Multimedia and its Elements

Multimedia means the use of **more than one type of media** – text, images, audio, video and animation – together in a **digital** form, usually with **user interaction**, on a computer or other device.

- Key features: multiple media, digital data, integration, **interactivity**.
- Example: an e-learning app that shows text, pictures, a video lesson, sound and a quiz.

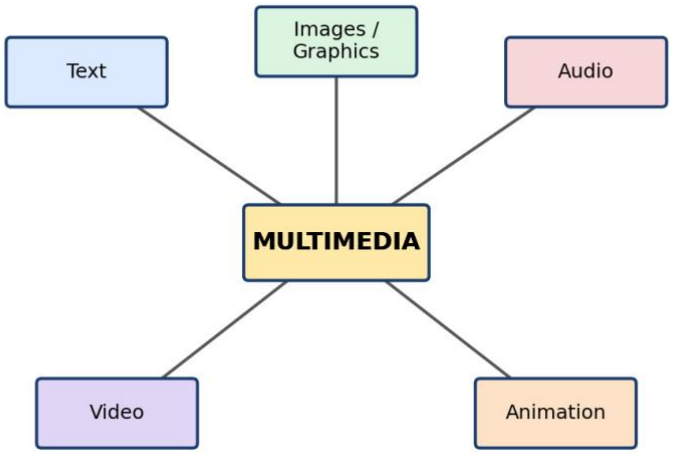


Fig. 1: Elements of multimedia

Element	Description	Common formats
Text	Words, headings, fonts, hyperlinks. Simplest and most used element.	TXT, DOC, PDF, HTML
Images / Graphics	Still pictures. Bitmap (pixels) or vector (shapes and lines).	JPEG, PNG, GIF, BMP, SVG
Audio	Sound: speech, music, effects. Digital audio is made by sampling.	WAV, MP3, AAC, MIDI
Video	Series of images (frames) shown fast, 25–30 frames per second, with sound.	MP4, AVI, MKV, MOV
Animation	Drawings or objects changed frame by frame to give motion (2D / 3D).	GIF, SWF, FLI

- **Interactivity** (menus, buttons, hyperlinks) is also treated as an important part of multimedia.

Q2. Applications of Multimedia

1. Education

- E-learning, online courses, smart classrooms and educational videos.
- Simulations and animations to explain difficult ideas (science, maths).
- Digital libraries, language learning apps, quizzes with instant feedback.

2. Entertainment

- Video games, movies, music and video streaming (YouTube, OTT apps).
- Special effects, animation films, virtual reality (VR) experiences.

3. Business

- Presentations, advertising, product demos and online catalogs.
- E-commerce websites, employee training and video conferencing.
- Information kiosks in malls, airports and banks.

4. Healthcare

- Medical imaging (X-ray, CT, MRI) stored and viewed digitally.
- **Telemedicine** – doctor consults a patient by video call.
- 3D models of the human body and surgery simulators for training.
- Patient education videos and health records with images.

5. Other fields

- Government (e-governance), tourism (virtual tours), journalism, engineering design and military training.

Q3. Architecture of a Multimedia System

A multimedia system is built in **layers**. Each layer uses the services of the layer below it. Data comes from **input devices**, is processed by the layers and sent to **output devices**.

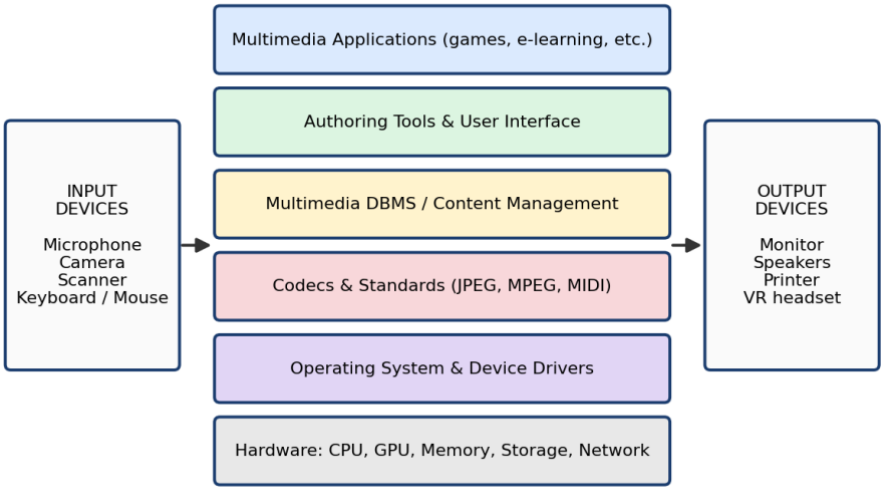


Fig. 2: Layered architecture of a multimedia system

Major components

- **Input devices** – microphone, camera, scanner, keyboard, mouse to capture data.
- **Hardware layer** – CPU, GPU, memory, storage and network card do the processing, storing and sending.
- **Operating system and drivers** – manage devices and give real-time scheduling for smooth audio and video.
- **Codecs and standards** – compress and decompress data (JPEG, MPEG, MIDI).
- **Multimedia DBMS / content management** – stores and retrieves media files.
- **Authoring tools and user interface** – used to create content and interact (menus, buttons).

- **Applications** – final programs like games, video players, e-learning.
- **Output devices** – monitor, speakers, printer, VR headset.
- **Network** – carries multimedia over LAN or the Internet (streaming, video calls).

Q4. Hardware and Software Requirements

Hardware requirements

Hardware	Why it is needed / examples
Processor	Fast multi-core CPU and GPU for decoding video, 3D and animation.
Memory (RAM)	Large RAM (8 GB or more) to handle big image and video files.
Storage	Large and fast storage – SSD / HDD, DVD, Blu-ray, pen drive, cloud.
Graphics card	For high-resolution display, 3D and video processing.
Sound card	Records and plays digital audio.
Display	High-resolution monitor (HD / 4K); speakers or headphones.
Input devices	Microphone, camera, scanner, graphics tablet, keyboard, mouse.
Network card	For streaming, downloading and video conferencing.

Software requirements

Software type	Examples
Operating system	Windows, Linux, macOS, Android
Media players	VLC, Windows Media Player
Image editing	Photoshop, GIMP, CorelDRAW
Audio editing	Audacity, Adobe Audition
Video editing	Premiere Pro, DaVinci Resolve
Animation / 3D	Blender, Maya, Adobe Animate
Authoring tools	PowerPoint, Adobe Director, HTML5 tools
Codecs and drivers	MP3, H.264, device drivers
Database	Multimedia DBMS to manage media files

Q5. Evolving Technologies in Multimedia

- **Virtual Reality (VR)** – user enters a fully computer-made 3D world using a headset (games, training).
- **Augmented Reality (AR)** – digital objects are shown over the real world through phone or glasses (Pokémon Go, furniture apps). **Mixed Reality** combines both.
- **High resolution displays** – 4K / 8K, HDR and OLED give sharper and more colourful pictures.
- **Streaming and cloud** – video and music played online without downloading (Netflix, YouTube, cloud gaming).
- **Better compression** – new codecs like HEVC (H.265) and AV1 give the same quality in smaller size.
- **5G networks** – very high speed and low delay for live video, VR and online gaming.
- **Artificial Intelligence** – automatic image/video generation, speech recognition, video recommendation, auto-subtitles.

- **3D and holography** – 3D movies, holographic displays and 3D printing.
- **Haptics and gesture control** – touch feedback and control by hand movements.
- **Wearables and IoT** – smart watches, smart TVs and connected devices play multimedia.

Q6. Multimedia Data Interface Standards

Standards are agreed rules that define **how multimedia data is formatted, compressed, stored and exchanged** between devices and software. Data interface standards allow different systems to understand each other.

- Examples: JPEG (images), MPEG (video), MP3 (audio), MIDI (music), H.264, HDMI, USB, RTP (streaming).

Types of standards

- **File / data format** – JPEG, PNG, MP4, WAV.
- **Compression** – JPEG, MPEG, MP3.
- **Hardware interface** – USB, HDMI, MIDI cable.
- **Network / streaming** – TCP/IP, RTP, RTSP.

Why standards are important

- **Compatibility** – a file made on one device plays on any other device.
- **Interoperability** between products of different companies.
- **Saves cost and time** – no need to build a new format for each product.
- Easy **exchange and sharing** of data over the Internet.
- Make **compression** possible, saving storage and bandwidth.
- Give **quality assurance** and long-term access to data.
- Help the industry grow because developers follow the same rules.

Q7. Role of JPEG, MPEG and MIDI

1. JPEG (Joint Photographic Experts Group)

- Standard for compressing **still images** (photographs); file extension .jpg / .jpeg.
- Mostly **lossy**: removes details the human eye cannot see easily. Compression about 10:1 to 20:1.
- Supports 24-bit colour (16 million colours).

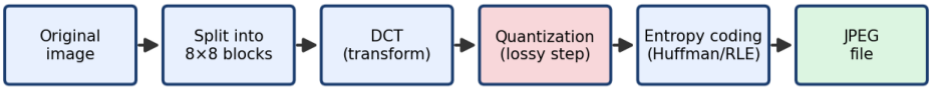


Fig. 3: JPEG compression steps

- **DCT** converts each 8×8 block into frequency values; **quantization** removes less important values (data loss); **entropy coding** shortens the final data.

2. MPEG (Moving Picture Experts Group)

- Standard for compressing **video and audio**.

- Uses **spatial compression** (like JPEG inside each frame) and **temporal compression** (only the changes between frames are stored, using motion compensation).

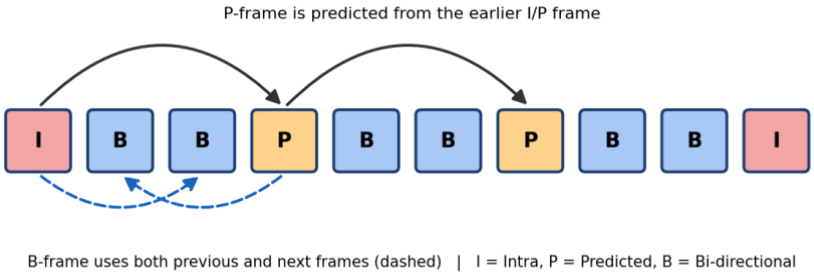


Fig. 4: MPEG frame types

- **I-frame** – complete picture. **P-frame** – predicted from earlier frame. **B-frame** – uses earlier and later frames (smallest).

Version	Use
MPEG-1	Video CD; includes MP3 audio
MPEG-2	DVD and digital TV
MPEG-4	Internet video, mobile, streaming (H.264)
MPEG-7 / 21	Description of media content / multimedia framework

3. MIDI (Musical Instrument Digital Interface)

- A **standard and protocol** for connecting musical instruments (keyboard, synthesizer) with computers.
- It does **not store sound**; it stores **instructions** like note on/off, pitch, speed (velocity), instrument and channel (16 channels).
- The sound is produced later by a synthesizer or sound card.
- Files (.mid) are **very small**; easy to edit notes and tempo; used for music composition and games.

Comparison

Standard	Media	Main role
JPEG	Still images	Compress photographs
MPEG	Video + audio	Compress moving pictures
MIDI	Music instructions	Control and exchange music data

Q8. Need for Data Compression and its Advantages

Data compression reduces the number of bits needed to store or send data. Multimedia files are very large when uncompressed.

Why is it needed? (examples)

Media	Calculation	Size
Image	1024 × 768 pixels × 3 bytes (24-bit)	≈ 2.25 MB
CD audio (1 min)	44,100 samples/s × 2 bytes × 2 channels × 60 s	≈ 10.6 MB
Video (1 min)	640 × 480 × 3 bytes × 30 fps × 60 s	≈ 1.66 GB

- Storage devices have **limited space**.
- Networks have **limited bandwidth** – large files take long to send.
- Video and audio must play in **real time** without breaks.
- Example: MP3 at 128 kbps is about 11 times smaller than CD audio (1411 kbps).

Advantages

- **Saves storage space** – more files in the same disk.
- **Faster transmission** – less time and data cost on the Internet.
- **Smooth streaming** and real-time playback even on slow networks.
- **Lower cost** of storage and bandwidth.
- Faster file copying, uploading and downloading.
- Makes multimedia on mobile phones and web possible.

Q9. Lossless vs Lossy Compression

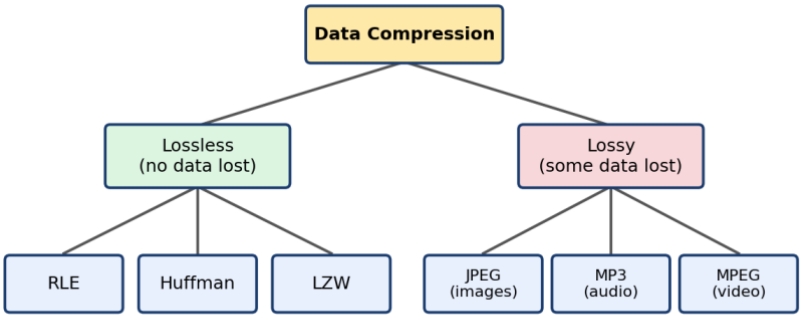


Fig. 5: Types of data compression

Lossless compression – the original data can be **fully recovered** after decompression.
Lossy compression – some data is **permanently removed**, so the original cannot be recovered exactly.

Point	Lossless	Lossy
Data loss	No loss; exact copy is restored.	Some data lost permanently.
Quality	Same as original.	Slightly lower than original.
Compression ratio	Low (about 2:1 to 3:1).	High (10:1 to 100:1).
File size	Larger.	Much smaller.
Used for	Text, programs, medical images, documents.	Photos, music, video, streaming.
Methods	RLE, Huffman coding, LZW.	DCT, quantization, perceptual coding.
Examples	ZIP, PNG, GIF, FLAC, BMP (RLE).	JPEG, MP3, AAC, MPEG / MP4.

Example of lossless: Run Length Encoding (RLE)

```
Original :  AAAABBBCCDAA   (12 characters)
Encoded  :  4A3B2C1D2A     (10 characters)
```

- Decoding gives back exactly **AAAABBBCCDAA** – nothing is lost.

- **Huffman coding** gives shorter codes to frequent symbols; **LZW** replaces repeated patterns with dictionary codes.

Example of lossy

- **JPEG** removes fine colour details; **MP3** removes sounds the ear cannot hear (very soft sounds near loud ones). The saved file is smaller but cannot return to the original.

Q10. Multimedia Database

A **multimedia database (MMDB)** is a database that stores, manages and retrieves **multimedia objects** – text, images, audio, video and graphics – along with their **descriptive data (metadata)**. It supports search by **content**, not only by name.

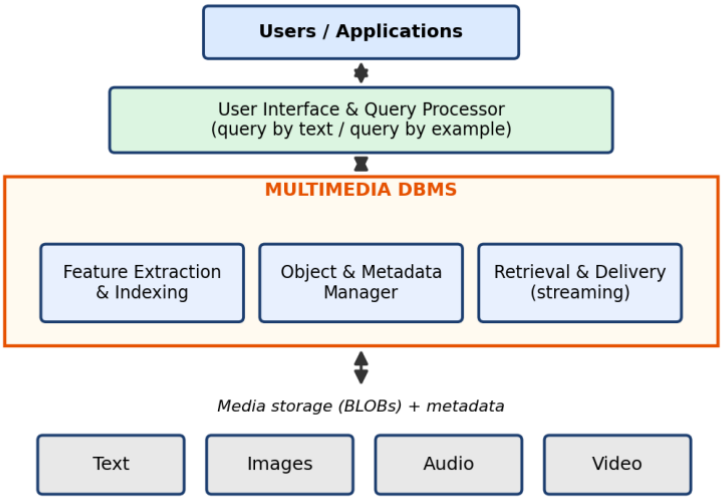


Fig. 6: Multimedia database architecture

Characteristics

- Stores **very large** objects (BLOBs – Binary Large Objects).
- Handles **many data types** in one system.
- **Content-based retrieval** – search by colour, shape, sound or by giving an example (query by example).
- Uses **metadata** (title, size, format, keywords) and special **indexing**.
- **Similarity search** – finds items that are alike, not only exact matches.
- Supports **time and space relations** and **synchronization** (audio with video).
- Needs **real-time delivery** (streaming) and high-speed storage.
- Provides security, copyright protection and multi-user access.

Components

- **Media storage** – holds text, images, audio and video files (BLOBs).
- **Metadata store** – details about each object.
- **Feature extraction and indexing** – finds features (colour, texture, pitch) and builds an index.
- **Object / database manager** – controls storage, updates and transactions.
- **Query processor and user interface** – accepts text and example queries.
- **Retrieval and delivery (streaming) module** – sends results to the user over the network.

Applications

- Digital libraries and museums; news and photo archives.
- Medical systems (X-ray / MRI image storage – PACS).
- Video-on-demand and online music services.
- E-commerce product catalogs (search by image).
- Security systems – face and fingerprint matching, CCTV archives.
- E-learning content storage and GIS (maps and satellite images).